

Innovative Approach to the Substation Design using Building Information Modeling (BIM) Technology

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Abstract—Rising power demand, on aging infrastructure, causes more and more issues, right up to blackouts. There is a national need for grid rebuilding of the power network over the next decade. The fact that large number of power engineers will be retired in the next years, forces the power industries to look for efficient design solutions. For this reason, it is important to achieve shortened project time, costs reduction, efficiency and accuracy, in substation design. Software which fulfills these requirements, using Building Information Modeling (BIM), is described in this paper. This technology provides planning, designing, building and managing 2D or 3D models, having the intelligence, geometry and data utilizing contained elements. Thanks to BIM, designers of different branches can work more collaborative. Software provides whole electro-mounting, designing, including civil and architectural review.

Index Terms--3D model, BIM, construction, substation design.

I. INTRODUCTION

Growing needs for power system development, for the purpose of electricity delivery, have been caused by industry development, increasing standards of living and population growth. The delivery of the electricity must raise the level of security and reliability but also raise the required quantity, quality and continuity [1]. There is an increasing need for construction of substations, which represent node points or transformer points of the network. However, substation design is a complicated process, which requires important analysis before the beginning of any design. It also requires finding the optimal solution, satisfying the number of restrictions and standards. Substation design is based on teamwork from different branches [2], [3].

The aim of this paper is to describe a software tool for substation design, based on Building Information Modeling (BIM) technology and will provide quick and logical designing [4], [5]. This software is about enhancing/improving the AutoCAD (CAD - Computer Aided Design) platform where all benefits and features of this platform are used with a standard engineering approach also with the advantage of a database hosted information model. This developed software is called PRIMTECH. The software uses common AutoCAD, AutoCAD Architecture or AutoCAD

MEP (Mechanical, Electrical and Plumbing Engineering) as the graphic editor. It embeds CAD - data in its database model via the specific HV - CAD Entities e.g. HV devices, wires, tubes, steel constructions, foundations, etc. The substation information model is created automatically in the database. Philosophy of the software is based on new technology for designing to become easier. All information of the model is supported using already created libraries (entis, project specific library and template library). Accent is on optimal placement of substation materials or elements disposition and utilize less of an occupied area [6]-[9]. Integrated calculation tools and several checkers can be used any time during the design process and thus help in making decisions or optimization in early phases. The software is developed according to international and national standards [10]-[20].

II. BIM TECHNOLOGY AND ITS IMPLEMENTATION IN SUBSTATION DESIGN

BIM is an intelligent process based model for planning, designing, building and managing of buildings and infrastructure. Comprehensiveness of BIM process is shown in Fig. 1. Objects that contain intelligence, geometry and data are used. If changes of some designing object are made, it is provided that these changes are reflected in every element of the design where this object is used. In this way, designers/users can work more collaboratively, constantly updating the model.

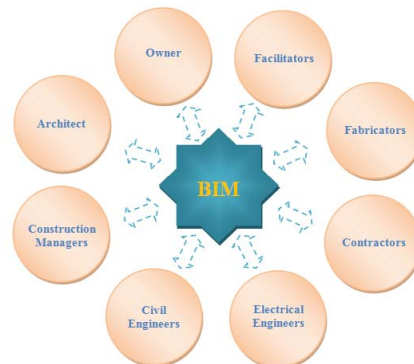


Figure 1. Comprehensiveness of BIM process

Model information is always present, with high reliability, so access to it is possible in any time and in any phase of the project [5]. Making decisions is improved through simulations and analyses. The technology used in this software allows the possibility of better coordination and communication between designers, of the different branches. BIM increases accuracy and reliability, while stimulating teamwork and developing design, from office to the real space.

PRIMTECH utilizes all the benefits of the BIM process and is an example of how this philosophy is applied to the electric substation design, where architects, civil engineers, mechanical engineers and electrical engineers work in coordination, with constant insight in to all information that this principle offers. The following features are important: 3D models with associated information (including structuring and grouping in systems, constructions, etc.) and providing drawings, estimations like bill of materials (BOM); quantities.

With this software it is easy to be efficient in the whole process from tender over implementation to the documentation, Fig. 2. The ability to provide offers to the one who announces the tender in a very short time is paramount and more importantly, that there is no information lost between the provider and the end-user. Offers that are given for the tender are therefore precise and this aspect is provided by BIM process.



Figure 2. Software workflow follows BIM philosophy

The software gives the entire 3D substation model in all its phases and in final integrated form, so there is a complete overview of planning and implementation. Output documentation (eg. layout drawings) has high quality because it provides insight in to all elements and details of substation.

III. SUBSTATION DESIGN METHODOLOGY

The core component of the software is the Model Manager, where folders are automatically created supporting the whole substation design process. Through this, the substation design methodology is defined, see Fig 3. The process is starts with a single line diagram, selecting a basic layout, from a template library, according to the functional design and available plots of land required [10]-[12]. Afterward, the engineering work is to bring this generic substation information model, step by step, to a complete specified information model, containing all detailed specifications for devices right down to clamps and screws. This integrated model is shown in Fig. 4. Substation designing consists of creating symbols (foundations and steel construction), assemblies (symbols and details) and modules (assemblies and connections). Undefined objects are used in the first stage of design (e.g. electrical devices, clamps, wires, etc.). This means these objects are universal objects of the exact type, but not the specific defined object. The software provides very simple object exchanges, with any objects, in any stage of the project.

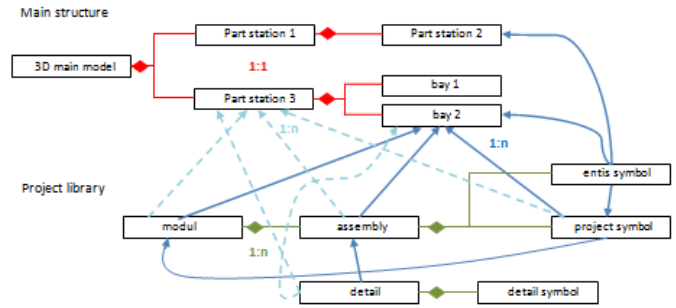


Figure 3. Project structure

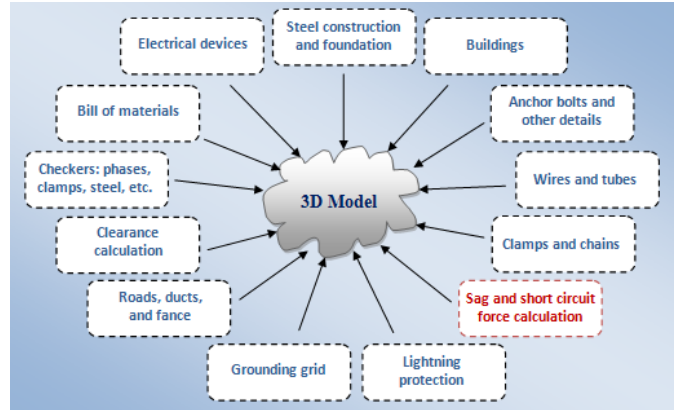


Figure 4. Integrated process 3D model

The library contains thousands of substation components, from different manufacturers, for electro-mounting. The component library ensures optimum performance (e.g. fast library item search and fast symbols displayed, even though the library is comprised of thousands elements within it). New designs of components, may be added so the users may extend their libraries. With the defining of all substation components, it is easy to make bills of all materials including the smallest of details such as anchors, bolts, contacts, etc. in order to define exact price or costs of the entire substation. An example of a 3D substation model of this software is shown in Fig 5.

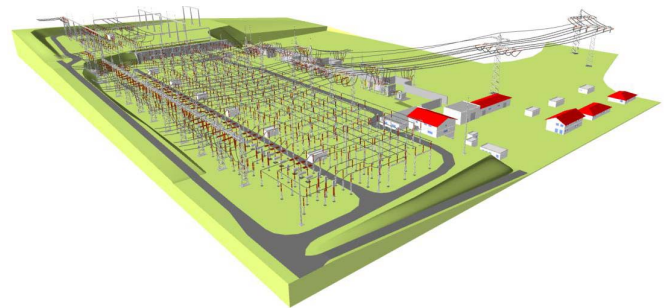


Figure 5. 3D substation model

IV. INTELLIGENT 3D OBJECTS

Most substation elements are intelligent 3D objects with linked information which are contained in entis, Fig. 6. This library contains undefined and defined objects (from different manufacturers), which are easy to adjust to the users'

requirements. Some intelligent objects are parametric foundations, steel structure, wires, tubes, etc.

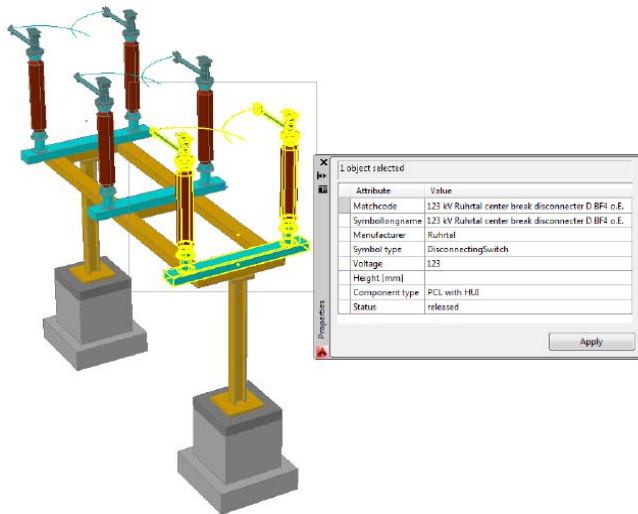


Figure 6. Intelligent 3D object - disconnector assembly

The entis libraries contains intelligent HV symbols that are exchangeable according to user's requirements. Symbols contain characteristic points for easier connection of objects in the substation, i.e. electrical, static and insertion.

New objects can be added to the library as shown in Fig. 7. The Model Manager component library is based on an Oracle or MS-SQL database, ensuring optimum performance even when the quantity of data is very large. Global servers are representing a company cloud for multisite use of library objects, as shown on Fig. 7.

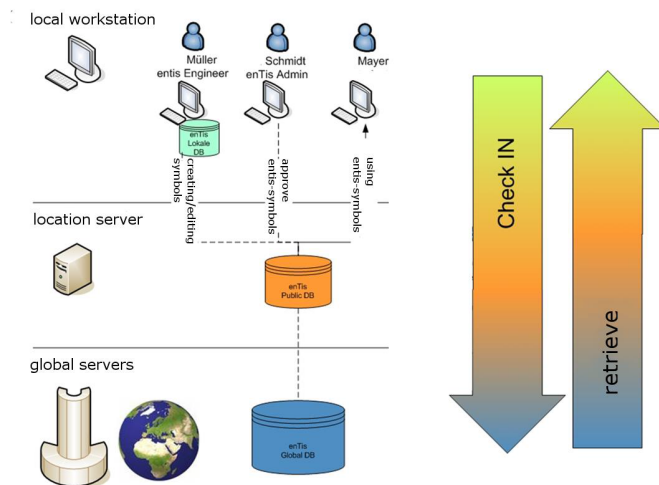


Figure 7. Database structure

V. DESIGN MODULES

A substation is saved in PRIMTECH as a substation information model, consisting of HV objects, embedded detailed 3D geometry, object associations and of metadata. In addition there are other intelligent entities, needed for full 3D

substation view, are developed in this software and are described below.

A. Field grids

Field grids, saved in the system according to voltage, serve as positioning aids for electrical equipment, and ensure that the distances between individual devices are not less than the entities needed for full 3D substation view, are developed in this software and are described below.

B. Steel structure

Steel construction tool is supported by a comprehensive 3D library including components such as steel lattice towers or supporting constructions for busbars. A comprehensive library of profiles is also available, with which fully customized steel girders can be generated.

C. Foundation

Foundation is parametric with flexible length, width, height and materials. Multiple geometries for single, double, square foundation head are supported.

D. Wires and tubes

Wires and tubes connections can be put together under the guidance of the connection configurator, and automatically generated in the 3D model. In addition to the wire, wire connections also include chains, springs and clamps, and the sag can be specified. Particular support is given to the three-phase design of an installation.

E. Cable ducts

Software supports the generation both of cast and of segmented cable ducts. Both the length and the volume of the cable ducts are available to assist the assessment of parts lists.

F. Roads, paths and fences

Convenient functions allow roads and paths, including crossings, to be generated quickly. The creation of fences with mesh wire or rigid fence segments is also supported. This can all be evaluated and then generated into a parts list automatically.

G. Earth cable network

PRIMTECH provides convincing functionality for processing grounding grids both for switchgear and buildings, e.g. for gas and air insulated substations [13]-[15]. The scope of the functions includes special tools for quick and easy generation of meshed networks with earth cable objects, as well as details for mesh nodes and device connections. The parts list evaluates multiple conductors and small parts, as well as, the vertical distance of cables to the devices without having to draw the vertical installation. Earth network components include earth connections, earth cables, earthing details, etc.

H. Building design

Libraries contains building components (walls, doors, windows, etc.) for control building creation.

I. Quantified Bill of Material (BOM)

The quantified BOM can be evaluated over the project, as a whole, or just over parts of substation. It can be used, for instance, to prepare a clamp list or if modifications are planned, to evaluate a newly designed part of the plant. Software calculates the volume (e.g. volume of concrete), the length (e.g. the length of wire or tube) and the area (e.g. area to be painted) available for evaluation. Bills of material can be created as output for Excel.

VI. CALCULATION MODULES

A. Lightning protection calculation

The software features a powerful lightning protection calculation, implementing conventional lightning protection calculation methods. Using lightning protection rods and wires as lightning arresters, visually illustrating the required results in the form of the lightning protection volumes and areas, Fig. 8.

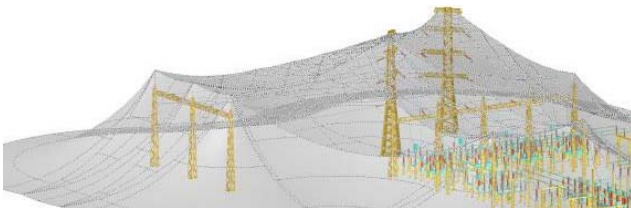


Figure 8. Lightning protection zone

Software supports lightning calculations using a rolling-sphere method. The sphere radius can be inserted freely according to international and national standards [16]-[18].

B. Phase checker

On the basis of the intelligent 3D models, the Phase Checker classifies all geometries into the following: live parts, earth parts, insulators, fences, streets.

All live parts are then classified, through automatic phase tracking, into all phases, two phases or each separately as well. As a side note, the phase checker can be also used to see if all electrical conductors are correctly connected, Fig. 9.

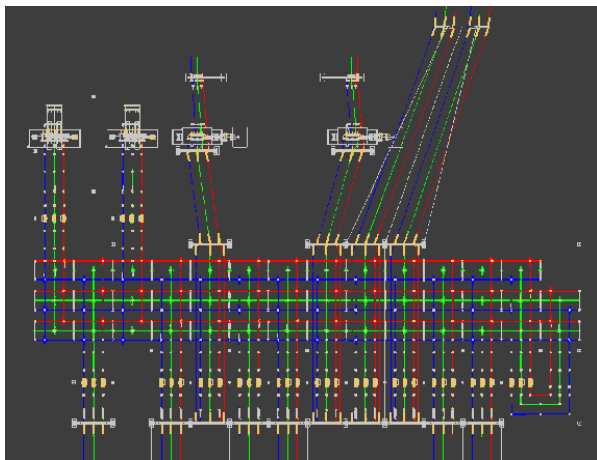


Figure 9. Phase checker

C. Clearance calculation

The construction of Air Insulated Substations (AIS) entails special requirements for the maintenance of minimum distances between the different phases, in live parts, as well as between live and earthed/grounded parts, while calculating the number of parts. Calculating clearances between live parts and surrounding fences or vehicles driving through the substation or even clearances between live parts and switched off parts. All relevant clearances in high voltage substations can be calculated easily and quickly.

In order to calculate clearances in a substation according to international and national standards [19], [20], there are two fundamental prerequisites that have to be fulfilled:

- 3D CAD-model of the substation;
- Information about each geometry - specifying if it is a live or earthed/grounded part, or an insulator or a special object e.g. fence or street. These requirements are fulfilled here, completely.

The following clearances can be calculated, Fig. 10.

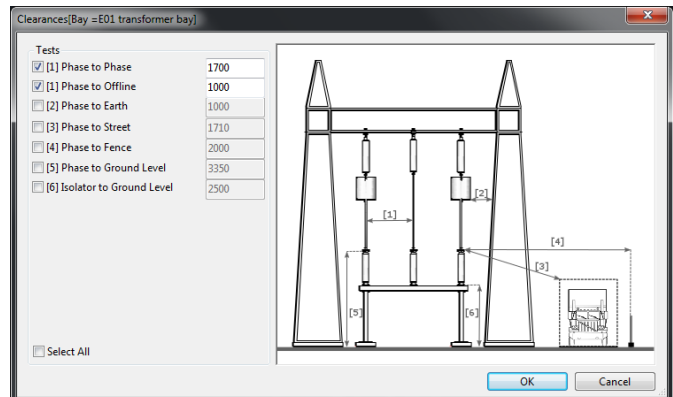


Figure 10. Input data for clearance checker calculation

All the required minimum distances throughout the whole substation can now be automatically determined based on the 3D model processed by the Phase Checker. The parts of the substation to be checked for sufficient clearance can be chosen individually and flexibly. The software will then visually highlight the areas which do not fulfill the minimum required distances to each other, Fig. 11.

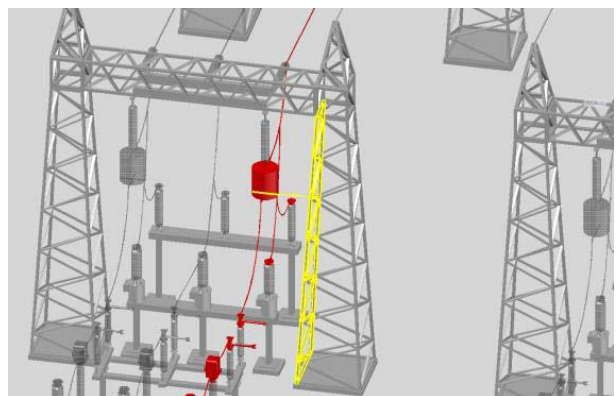


Figure 11. Clearance calculation

D. Substation elements checkers

There is also the possibility to check substation elements individually (clamps checker, string checker, insulator checker, etc.) which is very useful. This helps to check if all needed elements are there, or to check if some elements are missing.

It is well known that environment conditions like altitude, humidity, earthquake, wind or temperature have a big influence on substation design. As an example for wires, the sag is dependent on local climatic conditions (e.g. temperature, ice) or wind forces and short circuit forces, causing wires to swing. To avoid these problems after commissioning, simulation of these phenomena can be carried out. The calculation of the resulting tension or drop forces can also be done with software based on the substation information model.

VII. CONCLUSION

PRIMTECH has proved to be a very useful tool for substation design. Through the full view of a substation and the important aspects of it, the software provides quick, easy and efficient design. Simulations and adjustments give trustworthy substation view, before it is built, in real time and space. This offers interaction between designers of different branches, which is very important, during substation design.

The software provides engineering access. BIM process, which is the base of the software, is the most effective new approach in the world of design and provides solutions for plenty of issues that engineers deal with on each project, everyday. Substation design time is reduced considerably and the design process is simplified and logical.

This software is based on a database hosted BIM model, containing all information. It shows the complete model structure and it supports working (editing or exchanging) on that structure on CAD and database level. Other known software have implemented only CAD tools, so everything is file based, separating PRIMTECH as a very unique, reliable and powerful tool.

Classical 3D substation design software supports only two steps to concretize abstractions – functional design in a single line diagram and developing full specified design model in the next step. The single line diagram contains a small amount of needed information for the fundamental design decisions. Some of the information is not even available in that time (e.g. tensile forces). This kind of information first has to be developed through an iterative design process. In classical software this has to be made outside the CAD environment with several other tools and independent different data models. Optimizing the design in this workflow is very time-consuming. Often time constraints leads to more or less oversized and expensive designs or potentially poor design quality. The fundamental improvement of the PRIMTECH BIM process to classical 3D design is that the information model and the integrated calculations tools are highly specialized on supporting this processes of acquisition, managing and using information for better design decisions inside the design environment.

The software also contains integrated checkers: multiple scenario clearance check, clamp check, spacer check, etc. The complete set of checkers is unique, and separates PRIMTECH as unique as well. It also supports and controls the collaboration between several engineers e.g. physical design, civil design, building design within the same model in a multiple side environment. Power systems progress is followed by technology progress so it is important to use it in the best way. Modern 3D substation design, which is described in this paper credibly, shows that connection. The software further allows continuous upgrading and improvement according to users' and power system's requirements.

Sag and short circuit force calculation are the core features that are enabled in the newest versions of the software. It is the newest innovation that PRIMTECH provides. PRIMTECH is committed to growing newer versions and will continue to develop these, together with customers, in the future.

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